

Sports Profile - Endurance and Strength

ACTN3, ACE, PPARGC1A, AMPD1, NOS3, HIF1A, genomic scores of grip strength and fat-free mass, and loci load of force and lean mass

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

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1. Summary

Favourable. The genes of aerobic metabolism of the muscle are well: **PPARGC1A Gly482/Gly482** (C/C; PGC-1 α complete) and **AMPD1 normal** (G/G). Genomic grip strength is typical (52.4 and 51.3 percentiles).

Unfavourable. The mass free of genomic fat is below the median (27.0 and 29.5 percentiles) in the appendicular lean mass score of §27. The lean mass loci load is at percentile 11.1 and the grip strength in the 2.4 (very low, unlike the genomic force scores). NOS3 Glu298Asp is heterozygous (less endothelial nitric oxide).

Neutral. ACTN3 R577X is RX and ACE (marker I/D) is heterozygous, which gives a mixed fiber profile (nor "sprinter" or "marathonist" pure). HIF1A does not have the Ser582 variant.

What does it mean? The profile is "mixed, with good aerobic motor and low lean mass base". Genetics does not limit resistance or strength. The point to compensate is muscle mass, which tends to be genetically low. Therefore, progressive strength training and protein are the priority, as in §27, and aerobic training should respond well. The genes of muscle and bone are in §27; here is the sports profile.

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality $DR2 \geq 0.8$. Under this threshold SNP is treated as SNP not found in the data.
- **Key variants** 6 Selected SNPs from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 5 of the chip, 1 imputed and 0 not found.
- **Strand:** All genotypes are in the GRCh38 strand +. The published effect allele is compared with the observed alleles; 0 SNPs needed conversion to the complementary strand and 0 are palindromic (A/T or C/G), solved by the reference allele and the position. When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Polygenic scores (PGS):** 4 PGS Catalog scores calculated locally with 3,837,947 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$) calculated with exactly the same variants. Percentage ≥ 80 : MORE LIKELY/HIGH; ≤ 20 : LESS LIKELY/LOW; between 20 and 80: TYPICAL.
- **Loci load GWAS:** For each trait, all SNPs with $p \leq 5 \times 10^{-8}$ in GWAS Catalog (1 line for SNP, the most significant) are added as simple count of effect alleles (each allele valley 1). The sum of Sérgio F is compared with that of the reference population ($n = 633$) with exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium, so the load is a complement, not a substitute, of the polygenic scores. 2922 loci, 2586 with genotype.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

High percentile : strength or mass free of higher genetic fat (favourable). These scores are genomic (~1 million variants) and complement those of §27.

Muscle strength

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Handgrip strength (mean of the two hands)	PGS002143 (Privé 2022, LDpred2, 966 thousand variants)	946,066 966,283 (weighted coverage 97%)	52.4	TYPICAL	typical
Handgrip strength (mean of the two hands)	PGS003513 (Ding 2023, LDpred2-auto, 980 thousand variants)	972,627 979,739 (weighted coverage 99%)	51.3	TYPICAL	typical

Lean mass

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Fat free mass (whole body)	PGS002168 (Privé 2022, LDpred2, 967 thousand variants)	946,627 966,825 (weighted coverage 97%)	27.0	TYPICAL	typical
Fat free mass (whole body)	PGS003531 (Ding 2023, LDpred2-auto, 980 thousand variants)	972,627 979,739 (weighted coverage 99%)	29.5	TYPICAL	typical

4. Key variants by topic

Variants of genetics of sport. The individual effect is small and less replicated than in diseases; they serve to describe the profile, not to choose modalities.

Type of fiber and power

The type of fiber (fast vs slow) and the balance strength/power vs resistance depend in part on ACTN3 and the renin-angiotensin system (ACE). HIF1A adjusts to glycolysis.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
ACTN3	rs1815739	R577X	T/C	chip	T (1 copy/s)	neutral	R577X: the T allele (577X) does not produce alpha-actinin-3 in the fast fibers (II type). RR favors power and sprint; XX slightly favors resistance and is associated with more eccentric muscle damage (detail in §27)	Moderate
ACE	rs4343	I/D (proxy)	G/A	chip	G (1 copy/s)	neutral	Marker of polymorphism I/D: G allele follows D (plus ECA; associated with strength and power), A accompanies I (associated with endurance and muscle efficiency)	Weak
HIF1A	rs11549465	P582S	C/C	chip	T (0 copy/s)	without judgment	Pro582Ser: the T allele stabilizes HIF-1 α (more glycolysis; associated with potency and response to hypoxia)	Weak

Aerobic resistance and muscle metabolism

PGC-1 α (PPARGC1A) controls the mitochondrial biogenesis that aerobic training stimulates. AMPD1 maintains muscle energy during intense effort.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
PPARGC1A	rs8192678	G482S	C/C	chip	T (0 copy/s)	favourable	Gly482Ser: T allele (Ser) lowers the activity of PGC-1 α (mitochondrial biogenesis) and is associated with lower VO2max and resistance	Moderate
AMPD1	rs17602729	Q12X	G/G	chip	A (0 copy/s)	favourable	Q12X (C34T): allele A (X) inactive to muscle deaminase AMP (early fatigue and lower aerobic response to homozygous training)	Moderate

Oxygen and vessels

The endothelial nitric oxide (NOS3) dilates the vessels of the muscle in exercise.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
NOS3	rs1799983	E298D	T/G	imputation DR2=1.00	T (1 copy/s)	unfavourable (light)	Glu298Asp: T allele (Asp) gives less endothelial nitric oxide (vasodilation in exercise); associated with lower resistance performance	Weak

5. GWAS loci burden

Sum of all loci of GWAS Catalog ($p \leq 5 \times 10^{-8}$) of grip strength and lean mass, with alleles oriented towards higher value, compared to the reference population.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F effect alleles	Calibrated variants	Percentile	Judgement
Handgrip strength	642 (571)	567 of 1142	570	2.4	unfavourable
Lean mass/fat free	2280 (2015)	1921 of 4030	2012	11.1	unfavourable

6. Favourable

- **PPARGC1A Gly482Gly** (C/C): PGC-1 α with complete activity (mitochondrial biogenesis; associated with more VO2max and the best aerobic response). **Moderate.**
- **Normal AMPD1** (G/G): without early fatigue. **Moderate.**
- **Typical genomic grip strength**: percentiles 52.4 and 51.3 (the score of §27 was at the 30th percentile). **Moderate.**

7. Unfavourable

- **Genomic fat free mass below the median:** percentiles 27.0 and 29.5; lean mass loci load at 11.1. It is consistent with the appendicular lean mass at the 10th percentile (§27). **Moderate.**
- **Loci load of very low grip strength** (percentile 2.4), in contrast to the two typical genomic scores. The scores better consider the effects, but the loci signal reinforces that the force is not a strong base point. **Weak moderate.**
- **NOS3 Glu298Asp G/T:** a little less endothelial nitric oxide; small effect on performance. **Weak.**

8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Thin Mass Loss	Mass free of genomic fat at the 27.0 and 29.5 percentiles (as in §27)	Unfavourable
Loss of muscle strength	Typical genomic grip strength (52.4 and 51.3 percentiles)	Neutral
Sarcopenia	Low genetic lean mass; typical force; ACTN3 RX	Unfavourable (mild)
Oxidative Phosphorylation Decline	PPARGC1A Gly482Gly and normal AMPD1: good muscle mitochondrial motor	Favourable
Vascular Aging	NOS3 Glu298Asp heterozygous (less nitric oxide)	Unfavourable (mild)

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **Progressive strength training 3×/week as a priority**, with 2–3 series per muscle group, near the failure, and load to rise. It is the direct response to low genetic lean mass. *Suggestion (not applied)*.
2. **Include power** (fast movements with moderate load, low jumps, short sprints). Fast fibers are the first to lose with age, and the RX profile responds to both stimuli. *Suggestion (not applied)*.
3. **Aerobic zone 2 (2–3×/week) and 1 session of intervals/week**: the favourable PPARGC1A suggests a good mitochondrial response. *Suggestion (not applied)*.
4. **Sufficient and distributed protein** (~30–40 g per meal, 3–4 meals, with leucine ~3 g per meal), creatine monohydrate 3–5 g/day (as already suggested in §27). *Suggestion (not applied)*.
5. **Pre-training caffeine only in the morning and ≤ 200 mg/day (plan)**; not to train late with caffeine. *Suggestion (not applied)*.
6. **Measuring muscle strength and mass** (handgrip with dynamometer, body composition by DEXA or bioimpedance) 1×/year, to see if training compensates for genetics. *Suggestion (not applied)*.
7. **Omega-3 3.2–4 g/day (already in the plan)**: supports muscle protein synthesis; vitamin D3 of the plan. *Suggestion (not applied)*.

10. Limitations

- The classical genes of sport (ACTN3, ACE, PPARGC1A, AMPD1, NOS3, HIF1A) have small effects and inconsistent replication. They do not predict performance nor the ideal modality.
- The ACE I/D polymorphism is an insertion/deletion that the chip does not read directly; a marker (rs4343) was used.
- Strength and fat-free mass scores were trained in adults of the general population, not in athletes.
- The load of loci GWAS sums SNPs in linkage imbalance; it is descriptive.
- Conclusions only from genes; it is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F chip (GRCh38, strand +). Absent SNPs come from local imputation (Beagle 5.5, public panel of complete 30x genomes), used only with $DR2 \geq 0.8$. No genetic data came out of the analysis computer.
- **Alleles and strand:** The published allele of effect is compared with the alleles of the imputed VCF; in the palindromic SNPs the strand + (standard of the GWAS Catalog and the PGS Catalog) is assumed.
- **Polygenic scores:** Harmonized GRCh38 files from PGS Catalog (4 genomic scores: grip strength and fat-free mass). Score = Δ coefficient $\times$ dosage. The percentile is empirical in the reference population ($n = 633$), with the same variants.
- **Loci load:** GWAS Catalog ($p \leq 5 \times 10^{-8}$), alleles guided by the published effect signal, and empirical calibration in the same reference population.
- **Interpretation:** reference literature (Yang 2003 and MacArthur 2007 for ACTN3; Montgomery 1998 for ACE; Lucia 2005 for PPARGC1A; Rico-Sanz 2003 for AMPD1; Ahmetov 2016 for the revision of sports genes Jones 2021 for grip strength), GWAS Catalog and ClinVar. The complete references are available on request.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Type of fiber and power	ACTN3	rs1815739	T/C	chip	T	1	strand +	RX: mixed power/resistance profile (the most common)
Type of fiber and power	ACE	rs4343	G/A	chip	G	1	strand +	Probable ID: mixed profile
Type of fiber and power	HIF1A	rs11549465	C/C	chip	T	0	strand +	0 copy of the effect allele
Aerobic resistance and muscle metabolism	PPARGC1A	rs8192678	C/C	chip	T	0	strand +	without the risk allele
Aerobic resistance and muscle metabolism	AMPD1	rs17602729	G/G	chip	A	0	strand +	without the risk allele
Oxygen and vessels	NOS3	rs1799983	T/G	imputation DR2=1.00	T	1	strand +	1 copy of risk allele